

Problem

Solve the following integrodifferential equation

$$2\frac{dx}{dt} + 5x + 3 \int_0^t x dt + 4 = \sin 4t, \quad x(0) = 1$$

Step-by-step solution

Step 1 of 1

$$2x' + 5x + 3 \int_0^t x dt + 4 = \sin 4t; x(0) = 1$$

taking L.T

$$2[x(s) \cdot s - x(0)] + 5X(s) + \frac{3X(s)}{s} + \frac{4}{s} = \frac{4}{s^2 + 16}$$

$$\Rightarrow X(s) \left[2s + 5 + \frac{3}{s} \right] \frac{4}{s} - 2 = \frac{4}{s^2 + 16}$$

$$\Rightarrow X(s) \left[\frac{2s^2 + 5s + 3}{s} \right] + \frac{4 - 2s}{s} = \frac{4}{s^2 + 16}$$

$$= X(s)(2s^2 + 5s + 3) = \frac{4s}{s^2 + 16} + 2s - 4$$

$$= \frac{4s + 2s^2 + 32s - 4s^2 - 64}{s^2 + 16} = \frac{2s^3 - 4s^2 + 36s - 64}{s^2 + 16} = \frac{2(s^3 - 2s^2 + 18s - 32)}{s^2 + 16}$$

$$X(s) = \frac{2(s^3 - 2s^2 + 18s - 32)}{(s^2 + 16)(2s^2 + 5s + 3)} = \frac{As + B}{s^2 + 16} + \frac{Cs + D}{2s^2 + 5s + 3}$$

$$2(s^3 - 2s^2 + 18s - 32) = s^2(2A + C) + s^2(5A + 2B + D) + s(3A + 5B + 16C) + 3B + 16D$$

Comparing coefficients of different power of S,

$$2A + C = 2 \Rightarrow C = 2 - 2A$$

$$5A + 2B + D = -4$$

$$3A + 5B + 16C = 36$$

$$3B + 16D = -64 \Rightarrow B = \frac{-64 - 16D}{3}$$

$$\Rightarrow D = \frac{-64 - 3B}{16}$$

$$\text{using (1) in (3)} \quad 3A + 5B + 16(2 - 2A) = 36$$

$$32 - 32A = 36$$

$$\Rightarrow -29A + 5B = 4$$

$$\Rightarrow B = \frac{4 + 29A}{5} \quad \text{--- (5)}$$

$$5A + 2B - \frac{64 - 3B}{16} = -4$$

$$\Rightarrow 80A + 32B - 64 + 3B = -64$$

$$\Rightarrow 80A + 29B = 0$$

$$\text{Using (5) in above equation: } -80A + \frac{29}{5}(4 + 29A) = 0$$

$$400A + 29A = -116$$

$$400A + 841A = -116$$

$$A = \frac{-116}{1241}$$

$$5B = 4 + 29 \left(\frac{-116}{1241} \right) = \frac{4964 - 3364}{1241}$$

$$5B = \frac{1600}{1241} \Rightarrow B = \frac{320}{1241}$$

$$C = 2 - 2 \left(\frac{116}{1241} \right) = \frac{2482 - 232}{1241} = \frac{2250}{1241}$$

$$16D = -64 - 3B = -64 - 3 \left(\frac{320}{1241} \right) \\ = \frac{79424 - 960}{1241} = \frac{78464}{1241} \Rightarrow D = \frac{4904}{1241}$$

$$X(S) = \frac{1}{1241} \left[\frac{-116S + 320}{S^2 + 16} + \frac{2250 \left(S + \frac{5}{4} \right) + 2091.5}{2 \left[\left(S + \frac{5}{4} \right)^2 \left(\frac{1}{4} \right)^2 \right]} \right] \\ = \left[\frac{1}{1241} \left[-116 \cos 4t + 80 \sin 4t + e^{-1.25t} \left(1125 \cos \frac{ht}{4} + 4183 \sin \frac{ht}{4} \right) \right] \right]$$

[Comments \(1\)](#)



Anonymous

For everything up to $C = 2 - 2(116/1241)$ seems good. it should be negative - 116.1241 which would give you a higher value of c